

Indian Institute of Engineering Science and Technology, Shibpur  
BTech CST 4<sup>th</sup> Semester Mid-Semester Examination, February/March 2026

Programming Paradigms (CS2203)

Full Marks: 30

Time: 2 Hours

Attempt both question 1 and 2

1. Answer all questions

- a) Write a C++ class named Time to represent time in hours and minutes. With this class, the following operations shall be possible:

```
Time t1;           // Time 0 hours and 0 minutes
Time t2(10, 45);   // Time 10 hours and 45 minutes
Time t3(6, 30);    // Time 6 hours and 30 minutes
Time t4 = t2;      // A Time object created from an
                  // existing object
t4.display();      // Displays the time
Time T5 = t2+t3;   // Adds two time to instantiate a new one
```

- b) Suppose there is a C++ class named BankAccount as follows –

```
class BankAccount {
    int accNum;    // Bank account number
    double accBal; // Account balance

public:

    BankAccount(int num, double bal):
        accNum(num), accBal(bal) {}

    void display() const {
        cout << "Account Number: " << accNum << endl;
        cout << "Account Balance: " << accBal << endl;
    }
}
```

Declare and define a friend function named transferMoney(...) with suitable parameters to transfer money between two bank accounts.

(6 + 4 = 10)



2. Answer any 5 (not more than 5) from the following questions

- a) What do you understand by the terms 'state', 'behavior' and 'identity' of an object?
- b) Compare the efficiency of call by reference and call by address (value).
- c) What is Function Overloading or Static Polymorphism?
- d) When is it essential to implement a destructor in a class?
- e) What's the purpose of namespace in programming? Explain how namespaces are declared and used in C++ with an example.
- f) What do you understand by the term "Referential Transparency" in the context of functional programming?
- g) "Inline function speeds up execution and decreases the executable code size" – is this correct statement regarding inline function? – explain.
- h) What is Template Class? Explain using an example.

(5x4 = 20)